



Real-time ergonomic risk assessment in construction using a co-learning-powered 3D human pose estimation model

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Abstract

Work-related musculoskeletal disorders pose significant health risks to construction workers, making it essential to monitor their postures and identify physical exposure to mitigate these risks. This study presents a novel framework for real-time ergonomic risk assessment of workers in construction environments. Specifically, this study develops a lightweight human pose estimation (HPE) model with a residual log-likelihood estimation head and adopts pose-tracking technology to enable real-time recognition of workers' three-dimensional (3D) postures. In particular, this study proposes a novel co-learning method that enables the HPE model to learn two-dimensional (2D) and 3D features from multi-dimension datasets simultaneously, substantially enhancing the model's ability to capture 3D postures from 2D images. The proposed framework facilitates real-time ergonomic risk assessment, reducing potential risks to construction workers and offering promising practical applications.

1 | INTRODUCTION

Construction projects require intensive physical movement of workers' body parts, which results in potential work-related musculoskeletal disorders (WMSDs) in workers (Boschman et al., 2012). WMSDs are the main cause of workers' physical diseases, accounting for 18% of industrial accidents in Hong Kong (OSH, 2017) and 27.5% of nonfatal injuries in the United States (CPWR, 2018). WMSDs can pose a serious threat to workers' health and lead to a decline in production, thereby causing economic losses (Bernardes et al., 2019; Johansen et al., 2022). WMSDs are mainly due to the awkward postures of workers during construction (Keyserling et al., 1993). Thus, to avoid this type of potential injury, it is critical to monitor workers' postures and identify physical exposure to WMSDs.

The most widely adopted method to measure physical exposure to WMSDs is posture-based ergonomic assessment (Janowitz et al., 2006). The posture-based ergonomic assessment generally consists of two steps: posture estimation and risk assessment. The committed step is to accurately estimate workers' postures. With the maturity of wearable technology (B. Ren et al., 2022; Yoong et al., 2019), wearable sensing, especially initial measurement units (IMUs), is a promising alternative for precisely capturing human postures (Chen et al., 2017; D. Wang et al., 2015). Yan, Li, Li, & Zhang, 2017) proposed a real-time motion capturing and warning system based on wearable IMUs that enable workers to be aware of ergonomic risks and make adjustments for the prevention of WMSDs. Zhao and Obonyo (2020) used a lot of wearable IMU sensors to capture body keypoints and then adopted deep neural network models to recognize workers' postures.

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However, expensive whole-body motion-capture systems hinder the applicability of wearable sensing methods. In addition, it is inconvenient and uncomfortable for workers to wear sensors when executing construction projects (Yu et al., 2019). Consequently, wearable sensing methods are primarily used in laboratory environments (Yu et al., 2021) rather than in real construction projects.

With the rapid development of computer vision technologies, researchers have proposed many monitoring and assessment methods for civil and construction engineering (Guo et al., 2023). For example, a few researchers have used computer vision models to detect the equipment on construction sites and monitor the movement of equipment, based on which construction safety can be improved (Duan et al., 2022; Golparvar-Fard et al., 2013). Also, a few studies adopted computer vision technologies to conduct structure health monitoring (Bao et al., 2019; Ji et al., 2020). Other tasks, for example, construction project monitoring (Yan et al., 2023), scheduling (Adeli & Karim, 2001; Karim & Adeli, 1999a, 1999b), and management (Wu et al., 2023), can also be facilitated by computer vision technologies.

In recent years, a few researchers used computer vision technologies to estimate worker postures using two-dimensional (2D) images from monocular or depth cameras (He et al., 2016; J. Li et al., 2021; S. Ren et al., 2015). A straightforward method involves the direct classification of risky postures using 2D images (Martínez-Rojas et al., 2015; Seo & Lee, 2021). For example, Luo et al. (2019, 2020) combined deep learning and Bayesian learning to recognize workers' activities from 2D images. This end-to-end method can detect awkward postures from 2D images without additional operations. Nevertheless, it cannot recognize the specific postures of workers, for example, bending. As a result, it cannot quantitatively estimate ergonomic risks. In addition, a few studies (Harichandran et al., 2023; Jung et al., 2023) recognized construction risks based on construction equipment actions detected by deep learning models. Another commonly used method involves estimating ergonomic postures using 2D skeleton-based motion data captured by human pose estimation (HPE) models that adopt a range camera and OpenNI middleware to estimate the worker posture (Newell et al., 2016; Ray & Teizer, 2012; Xiao et al., 2018). Roberts et al. (2020) estimated and tracked 2D worker poses using an HPE model and then predicted worker activity classes of each frame based on the input of red-green-blue (RGB) video of a worker in construction environments. Jingluan Wang et al. (2021) developed a posture detector based on an HPE model that could identify the 2D limb angles and state of workers in working videos. There are other studies regarding the identification of worker postures from 2D images using HPE models (Yan, Li, Wang, et al., 2017; Zhao & Obonyo, 2021), details of which are not presented in this section for brevity.

However, there are significant errors in recognizing the three-dimensional (3D) postures of workers with 2D skeleton-based motion data because 2D skeleton-based motion can differ significantly from 3D skeleton-based motion. Hence, 3D skeleton-based motion data should be obtained to precisely recognize the postures of the workers. Tang and Golparvar-Fa (2021) first captured 2D body keypoints using an HPE model and then adopted an off-the-shelf keypoint conversion model to convert the 2D body keypoints to 3D body keypoints. Similarly, Zhang et al. (2018) developed a multi-stage convolutional neural network (CNN) model for lifting 3D keypoint locations from 2D images. Specifically, the first three stages of the model were designed to predict the 2D keypoints, and the remaining were used to lift the 3D keypoint locations from the predicted 2D keypoints. However, it is difficult to directly capture 3D keypoints (i.e., the 3D skeleton) from 2D keypoints, given that one 2D skeleton can correspond to various 3D poses (C. Li & Lee, 2019). In addition, compared with a single-stage model, that is, directly predicting 3D keypoint locations from 2D images, using two models or a multi-stage model to capture 3D skeletons may significantly increase the complexity of the HPE model, thereby decreasing the efficiency of the ergonomic assessment system in practical applications.

It is noteworthy that the ergonomic risk assessment of workers needs to be real time to make sense given that a post-event assessment cannot contribute to the worker's timely adjustment of awkward postures during construction. However, most of the aforementioned vision-based methods adopt heatmap-based HPE models that are too heavy to recognize the 3D postures of workers in real time (Roberts et al., 2020; Yu et al., 2021). Moreover, existing studies generally utilize an object detection model (e.g., Fast region-based convolutional neural network (Fast R-CNN); Girshick, 2015) to detect the worker pixel in each RGB frame of the videos. This in turn decelerates the ergonomic assessment because it is computationally expensive to run the adopted object detection models for all frames of working videos.

To address the abovementioned problems, a framework that can assess the ergonomic risks of workers in real time using a regression-based 3D HPE model powered by the co-learning strategy is proposed. The main contributions of this study can be summarized as follows.

1. This study proposes a novel co-learning method that enables the HPE model to simultaneously learn human pose information from both 2D and 3D datasets. This method significantly enhances the HPE model's ability to recognize 3D postures of workers directly from 2D images.
2. This study develops a 3D regression-based HPE model that accurately recognizes the 3D postures of a targeted

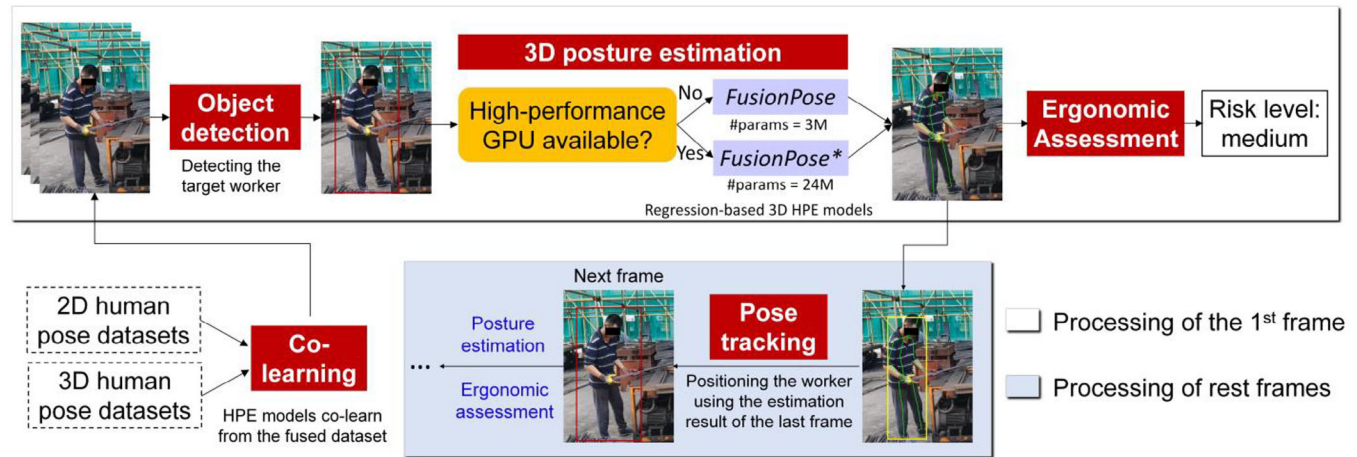


FIGURE 1 Framework of real-time ergonomic risk assessment in construction. HPE, human pose estimation. #params, number of parameters.

worker from 2D images in real time. In addition, this study leverages pose-tracking technology to directly detect the target worker based on the recognition results of the previous frame, eliminating the need for a separate detection model for all frames. The lightweight HPE model and pose-tracking technology together enable real-time recognition of 3D postures from 2D images.

3. The proposed framework can achieve real-time recognition of the 3D postures of workers in each frame of videos, based on which the ergonomic risks of workers can be effectively and efficiently quantified and assessed, holding great promise for practical utilization in construction projects.

The remainder of this paper is organized as follows. Section 2 briefly describes the proposed framework, while Section 3 introduces the corresponding underlying methodologies in detail. Section 4 illustrates the training and validation datasets and corresponding results of the regression-based 3D HPE model. The applicability of the proposed method is illustrated using a case study in Section 5. In Section 6, the advantages, limitations, and potential applications of the proposed framework are discussed. Section 7 gives the main findings of this study.

2 | FRAMEWORK

The proposed framework is shown in Figure 1. Specifically, a novel co-learning method that integrates multi-source data to train the 3D HPE model is first proposed to decrease the ambiguity from 2D images to 3D skeletons, which therefore improves the generalization and accuracy of the 3D HPE model. Co-learning indicates that 2D and

3D human pose datasets are first integrated into a fused dataset, in which the label format of all images is uniform, and then the HPE models co-learn 2D and 3D human postures from the fused dataset with the same loss function. The fused dataset contains both diverse 2D human pose information (i.e., pixel coordinates) from 2D datasets and depth information from 3D datasets. Therefore, the HPE model's ability to capture the pixel coordinates of workers can be trained sufficiently based on the fused dataset, which also enhances the HPE model's ability to predict depth coordinates. Refer to Section 3.1 for more details on the co-learning method. Then, an object detection model is used to detect the target worker in the first frame. The detected result provides the input of the developed regression-based 3D HPE model. The lightweight model is modified on the model proposed by J. Li et al. (2021), which can directly capture 3D keypoint positions, that is, 3D postures of workers, from 2D images in real time. In particular, two models with different sizes, that is, *FusionPose* with around 3 M parameters and *FusionPose** with around 24 M parameters, are provided for different computational resources. Specifically, if high-performance computational resources (e.g., Nvidia RTX 3090 Graphic Processing Unit (GPU)) are available, *FusionPose** can be used to obtain more accurate results; while *FusionPose* is a good alternative that can realize comparable performances on ordinary devices (e.g., laptop GPU). Refer to Section 3.2 for details of the two models. Subsequently, the pose-tracking technology (Bazarevsky et al., 2020) that uses the keypoint detection results of the last frame to detect the target worker in the current frame is introduced to this study (see Section 3.3), making it unnecessary to run the object detection model for all frames, which drastically improves the efficiency of the worker posture estimation in practical applications. Finally, the most widely used postural



analysis tool, Rapid Entire Body Assessment (REBA; Hignett & McAtamney, 2000), is adopted to quantify ergonomic risks based on the obtained 3D postures of workers (see Section 3.4).

Trained on the fused datasets, the modified regression-based 3D HPE models, *FusionPose**, realizes the state-of-the-art performance (i.e., 47.5 mean per joint position error [MPJPE]) on the Human3.6 M validation set (Ionescu et al., 2013), with 0.2-mm improvement and much lower computation cost. *FusionPose* realizes comparable performance to *FusionPose** while reducing computation cost by 93.7% (see Section 4.3.2). These results demonstrate the high accuracy of the proposed framework in capturing 3D keypoint positions from 2D images. To further verify the accuracy and generalization of the models, a new validation dataset is established by the authors for construction workers. Models *FusionPose* and *FusionPose** realize 76.0 and 82.8 mean average precision (*mAP*) on the validation dataset, respectively, both of which are greater than 70.0, which is usually regarded as high precision (see Section 4.3.4). Finally, the proposed framework is applied to assess the ergonomic risks of the workers in videos recorded in real construction projects. The results indicate that the proposed framework can accurately quantify the ergonomic risks of a frame within an average of 17 and 32 ms with an Nvidia RTX 3090 GPU, while 37 and 78 ms with a 6G laptop GPU with *FusionPose* and *FusionPose**, respectively (see Section 5).

Details of the underlying methodologies used in each part of the framework, that is, (1) co-learning from the fused dataset, (2) model, (3) pose tracking, and (4) HPE and ergonomic risk assessment, will be presented in Section 3.

3 | METHODOLOGY

3.1 | Co-learning from fused dataset

A novel co-learning method is proposed to improve the accuracy of the HPE model in capturing the 3D keypoint positions of workers directly from 2D images, with learning 2D and 3D features simultaneously from the fused dataset that integrates multi-source data.

Compared with 3D data, 2D data are much easier to be annotated (Yu et al., 2021). As a result, the open-source 2D human pose datasets are significantly higher in number and more diverse in the scene than 3D human pose datasets. Hence, the HPE models trained on 2D datasets generally perform better than those trained on 3D datasets. Specifically, HPE models pre-trained on 2D datasets can accurately capture the 2D postures of workers but without depth information. On the other hand, although 3D human pose datasets enable HPE models to capture the 3D

postures of workers, pre-trained 3D HPE models generally exhibit low generalizations due to the similarity between different 3D human pose datasets. This in turn leads to low accuracy of HPE models trained on 3D human pose datasets in practical applications. Therefore, a co-learning method is proposed to integrate 2D and 3D human pose datasets, which enables the HPE model to co-learn 2D and 3D features from the fused dataset, enhancing the HPE model's ability to accurately recognize the 3D postures of workers in practical applications.

The proposed co-learning method combines the strengths of two distinct types of datasets, that is, the diversity of 2D datasets, which encompass a wide range of scenes, and the completeness of 3D datasets, which include depth information. By integrating the two types of datasets, the proposed method enables the HPE model to accurately capture the 2D postures of workers, which serves as a foundation for predicting depth coordinates, thereby enhancing the accuracy of depth coordinate predictions. In addition, training the HPE models sequentially on a single dataset may lead to biased models, as the loss functions along with the optimization directions may vary between different datasets. For example, the HPE models trained on 3D datasets may have poor generalization, as the optimization direction is to fit similar human postures in indoor scenes. While being trained on 2D datasets, since the optimization direction is to fit 2D human postures, the HPE models' ability to capture depth information may decrease. However, the proposed co-learning method can address the above issue by training the HPE models on 2D and 3D datasets simultaneously with the same optimization direction, improving the prediction accuracy and generalization of the models.

Given that the body keypoint scheme varies between different human pose datasets, the key point of the proposed co-learning method is to integrate 2D-pixel coordinates from diverse 2D human pose datasets and 3D coordinates including depth information from 3D human pose datasets. Therefore, the body keypoints in several datasets are first merged to obtain a fused dataset. As shown in Figure 2, 2D human pose datasets, Max Planck Institute for Informatics (MPII) (Andriluka et al., 2014) and Common Objects in Context (COCO) (Lin et al., 2014), annotate 16 and 17 body keypoints, respectively, while 3D human pose datasets, Human3.6 M (Ionescu et al., 2013) and UP3D (Lassner et al., 2017), annotate 18 and 24 body keypoints, respectively. The fused dataset covers all annotated keypoints in the abovementioned 2D and 3D datasets by merging the same keypoints in different datasets as one keypoint. For instance, Keypoint 6 in MPII and Keypoint 0 in Human3.6 M and UP3D are merged as Keypoint 0 in the fused dataset. Consequently, a new body keypoint scheme that contains 32 keypoints is generated. Furthermore, the

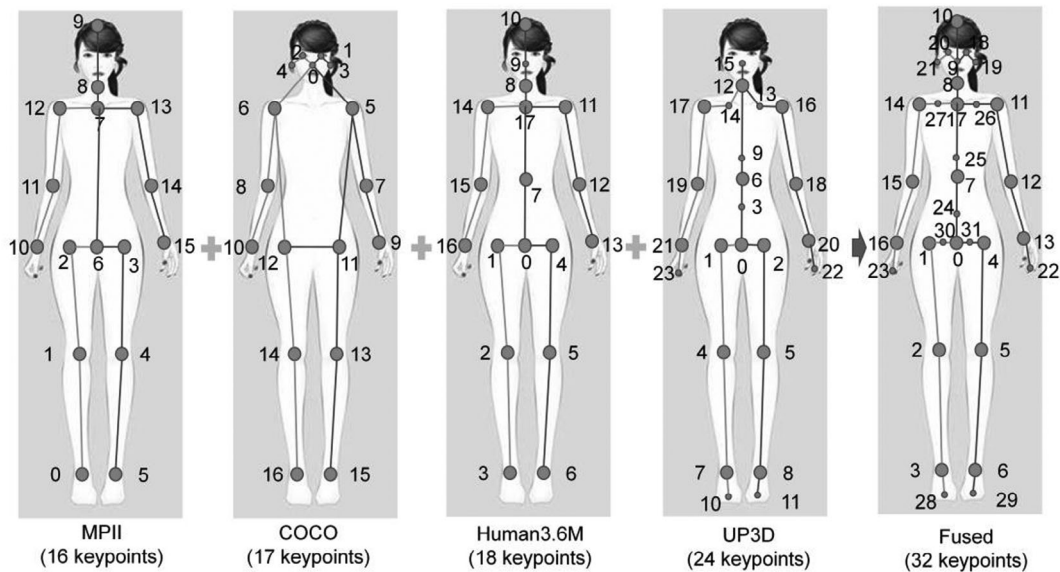


FIGURE 2 Schematic diagram of body keypoint fusion of different datasets.

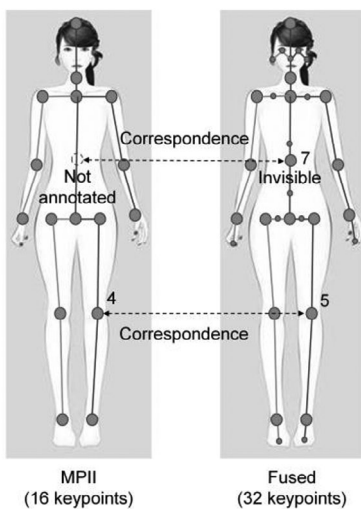


FIGURE 3 Schematic diagram of unifying annotation format of body keypoints in the fused dataset.

annotation format of the body keypoints is unified in the fused dataset that will be used to train the 3D HPE models. Specifically, as shown in Figure 3, the one-and-one correspondence between body keypoints in the original and fused dataset is first established, for example, Keypoint 5 in the fused dataset corresponds to Keypoint 4 in MPII. Moreover, the visibility of the body keypoints that are not annotated in the original dataset, but should be annotated in the fused dataset, is labeled as invisible, for example, Keypoint 7 in the fused dataset is labeled as invisible with correspondence to MPII in Figure 3. Finally, the visibility of the depth coordinates of body keypoints in the 2D datasets is marked as invisible, for example, the depth coordinates of all body keypoints in MPII should be set as invisible in Figure 3.

If a coordinate of a keypoint is marked as invisible, the loss function does not calculate the error of this dimension of the keypoint. In other words, the HPE models learn nothing from this dimension of the keypoint in the corresponding image, and vice versa. It should be noted that a keypoint labeled as invisible in the fused dataset represents this keypoint is not annotated in the original dataset rather than it is invisible or occluded in real-world scenes. Taking Keypoint 0 in the fused dataset as an example, the HPE models trained on the fused dataset can learn the 2D-pixel coordinates of this keypoint from images corresponding to MPII (annotated as Keypoint 6) and 3D coordinates from images corresponding to Human3.6 M and UP3D (annotated as Keypoint 0). However, images in COCO provide nothing for the HPE models to learn about this position, as this keypoint is not annotated in COCO (annotated as invisible in the fused dataset correspondingly). In general, all open-source human pose datasets cover 12 keypoints for four limbs. Hence, the fused dataset ensures that the HPE models can learn sufficient information about the main skeleton (two legs and two arms) of people in 2D images, which is significant for ergonomic risk assessment.

Hence, trained on the fused dataset, the HPE models can learn 2D information from diverse 2D human pose datasets as well as depth information from 3D human pose datasets, which enables HPE models to accurately capture 2D-pixel coordinates and comparatively accurately capture depth coordinates of body keypoints from 2D images. When the fused dataset is used to train the HPE models, the loss is calculated as follows:

$$L = \sum_{j=1}^J \sum_{l=1}^3 x_i^j w_i^j \quad (1)$$

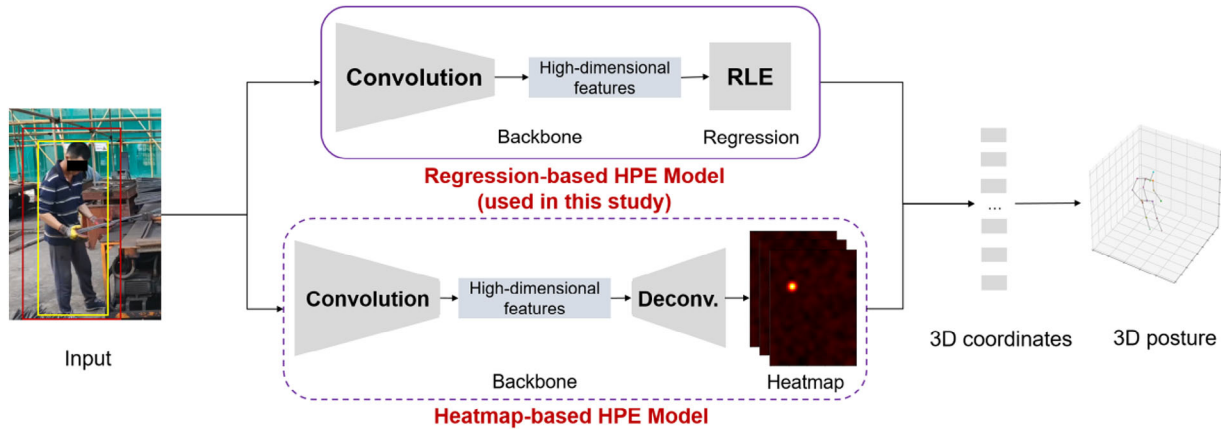


FIGURE 4 Illustration of three-dimensional (3D) posture estimation. RLE, residual log-likelihood estimation. Deconv., Deconvolution.

where x_i^j denotes the error between the prediction and annotation of i th coordinate of j th body keypoint, w_i^j denotes the weight of i th coordinate of j th body keypoint, and J denotes the number of body keypoints in the fused dataset. If the coordinate is marked as invisible, w_i^j equals 0 and otherwise equals 1.

3.2 | Model

In this study, the adopted two types of models, that is, the object detection model and the regression-based 3D HPE model, are used to capture the 3D postures of workers directly from 2D images.

Note that the size of the input to the HPE model is small (e.g., 256×192 pixels). As a result, the worker in the input image may also be small, making it necessary to detect the worker in the image first and then use the scaled image of the worker detected as the input to the 3D HPE model. Otherwise, the worker in the image may be so small that the HPE model cannot accurately capture the 3D posture of the worker. Therefore, Faster R-CNN (S. Ren et al., 2015), a mature object detection model that has been widely used in different object detection tasks, is adopted to detect workers in the first frame of the video. Remind that in the proposed method, the object detection model is only needed for the first frame of the video, and the worker in each of the subsequent frames is automatically detected by the keypoint recognition result of the last frame. The pre-trained Faster R-CNN can accurately detect persons, buses, and stores, among other things. Nevertheless, only the person detection results are needed in this study. It should be noted that the detected box (yellow box in Figure 4) contains only the target worker, which cannot be well-recognized by the HPE model, as it almost contains no background information. Thus, a scaling factor greater than 1 (e.g., 1.25) is utilized to expand the detected

box to obtain the input (red box in Figure 4) for the HPE model.

Generally, there are two types of HPE models: heatmap- and regression-based HPE models. As shown in Figure 4, heatmap-based HPE models first extract high-dimensional features from the input through convolutions, based on which deconvolutions are used to generate heatmaps of body keypoints. In each heatmap, the hottest point represents the position of a keypoint that can be calculated back to the original coordinates as well as 3D postures. Heatmap-based methods can accurately capture the positions of keypoints as they estimate the probability distributions of keypoints. Correspondingly, heatmap-based methods require high computational costs and storage demands (J. Li et al., 2021; Jinbao Wang et al., 2021). In contrast, regression-based HPE models directly estimate the coordinates of keypoints based on the extracted high-dimensional features, which is much more efficient for HPE tasks. Recently, many studies attempted to improve the capacity of regression-based methods and realized significant success (J. Li et al., 2021; Y. Li et al., 2022; Mao et al., 2022). In this context, the regression-based HPE models with residual log-likelihood estimation (RLE) that can realize comparable performance with heatmap-based models are adopted in this study, which exhausts significantly fewer computation and storage resources.

Regarding the backbone, most of the existing studies adopted heavy models, such as Residual Networks (ResNets) (He et al., 2016) and Vision Transformer (ViT) (Dosovitskiy et al., 2021), to realize accurate results. However, this study develops a significantly lighter model by substituting the standard convolutions in the backbone with depthwise separable convolutions. This lightweight model can realize real-time recognition of the 3D postures of workers even though no high-performance GPU is available, significantly expanding the application prospects of

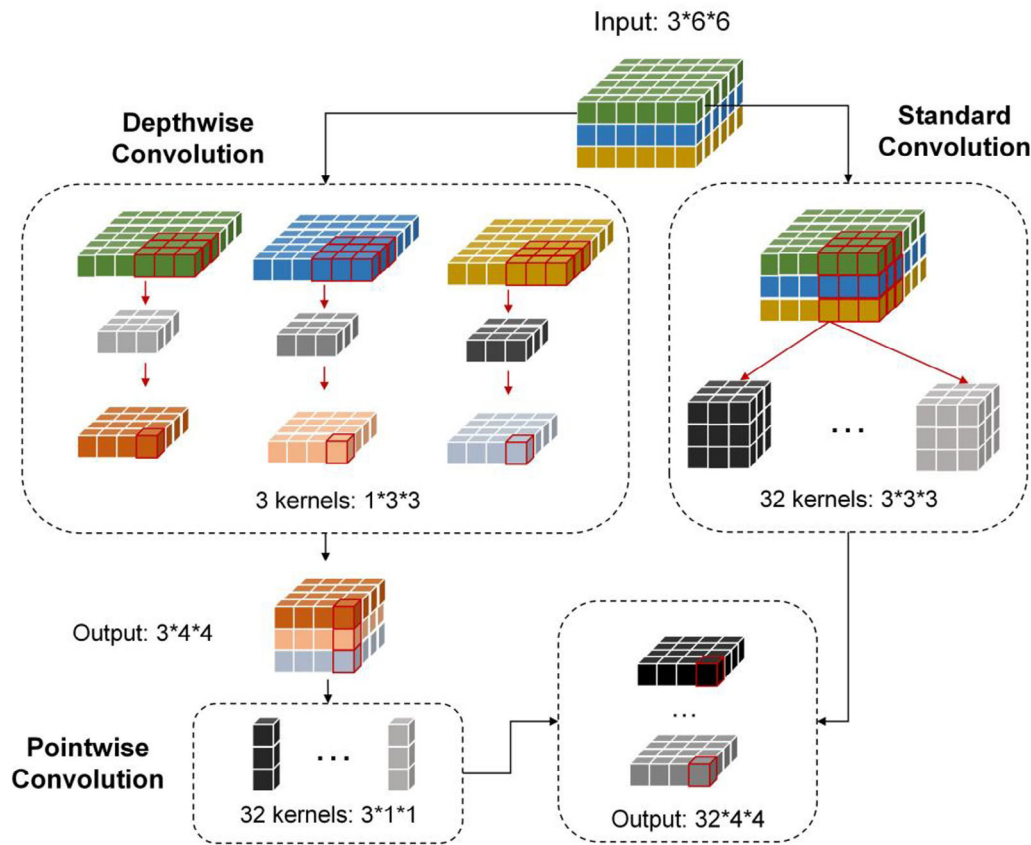


FIGURE 5 Comparison between standard convolutions and depthwise separable convolutions.

the proposed method. As shown in Figure 5, standard convolutions are widely used in many CNNs, while separable convolutions factorize a standard convolution into a depthwise convolution and a 1×1 convolution named a pointwise convolution (Howard et al., 2017). Specifically, the depthwise convolutions apply a single filter kernel to each input channel, for example, three kernels with the same size are used in Figure 5, individually extracting features of each channel. The pointwise convolutions then apply 1×1 convolutions to combine the separated features, and each 1×1 filter generates each output channel. It is important to note that depthwise convolutions maintain the same number of input channels while potentially altering the input size. For instance, as illustrated in Figure 5, the depthwise convolutions transition the input size from 6×6 to 4×4 . In contrast, pointwise convolutions preserve the input size but can modify the number of channels. As shown in Figure 5, pointwise convolutions transform the number of channels from 3 to 32.

According to Sandler et al. (2018), depthwise separable convolutions can realize comparable performance with a similar standard convolutional architecture with approximately one-tenth of computational costs. For example, as shown in Figure 5, the input size and output size are $3 \times 6 \times 6$ and $32 \times 6 \times 6$, respectively. The computational costs (i.e.,

the number of multiplication) of standard convolutions with $3 \times 3 \times 3$ kernels are:

$$3 \times 3 \times 3 \times 4 \times 4 \times 32 = 13,824 \quad (2)$$

However, the computational costs of depthwise convolutions are only:

$$1 \times 3 \times 3 \times 4 \times 4 \times 3 + 3 \times 1 \times 1 \times 4 \times 4 \times 32 = 1968 \quad (3)$$

Therefore, two typical CNNs, that is, MobileNetV2 (Sandler et al., 2018) and Resnet50 (He et al., 2016), which, respectively, consist of depthwise separable convolutions and standard convolutions, are adopted as the backbone of the regression-based HPE model. For brevity, the modified regression-based 3D HPE models are named *FusionPose* (corresponding to MobileNetV2) and *FusionPose** (corresponding to Resnet50) hereinafter.

The procedure for recognizing the 3D posture of a worker using *FusionPose* or *FusionPose** is shown in Figure 4. The detected target worker (obtained by the object detection model or detection result of the last frame) is used as the input for the model. The backbone first captures the high-dimensional features of the input.

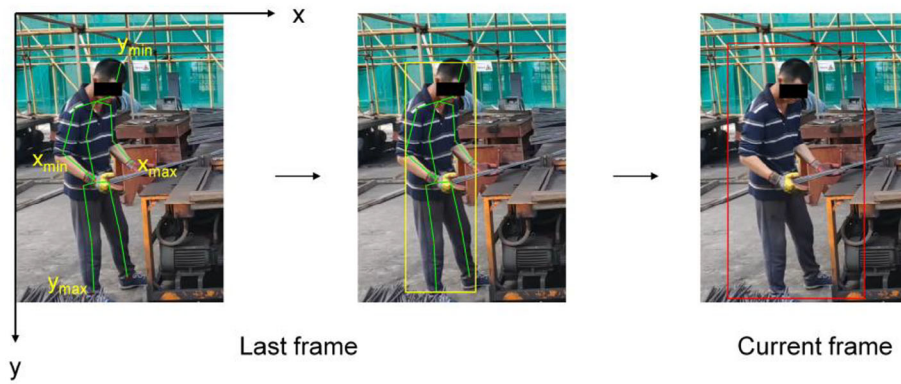


FIGURE 6 Illustration of pose tracking for all frames.

Subsequently, the RLE directly returns the 3D coordinates of all body keypoints to generate a 3D worker posture.

3.3 | Pose tracking

The pose-tracking technology, proposed by Bazarevsky et al. (2020), is adapted to detect the worker based on the detection results of the last frame. To the best knowledge of the authors, this study is the first to adopt the pose-tracking technology to accelerate the assessment of ergonomic risks to workers. Considering that the worker's position in the current frame is similar to that in the last frame for work posture estimation tasks, the keypoint detection result of the last frame can be used to detect the target worker in the current frame. Specifically, as shown in Figure 6, the maximal and minimal pixel coordinates among all body keypoints are first calculated based on the detection results in the last frame. Subsequently, an approximation bounding box of the target worker in the image can be easily obtained (yellow box in Figure 6). Then, a scaling factor greater than 1 is utilized to expand the box area to cover the entire worker in the image in the current frame (red box in Figure 6). Evidently, the expanded area provides more comprehensive information for the subsequent HPE model, enabling the model to accurately capture the 3D posture of the worker.

It is noteworthy that in practical applications, it is still necessary to use the object detection model periodically to detect the target worker to avoid errors due to unexpected situations like the temporary disappearance of the target worker in the field of the camera. Besides, there can be more than one worker in the video. For this circumstance, all workers should be identified first and then recognize the posture of each worker. Nevertheless, this is beyond the scope of this study and will be addressed in future research.

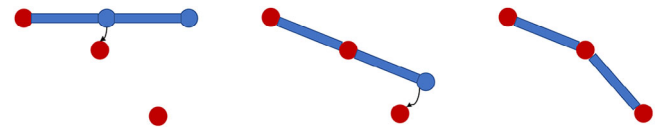


FIGURE 7 Schematic diagram of forward kinematics. Blue denotes the initial skeleton and red denotes the detected body keypoints of the current frame.

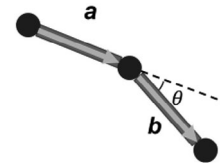


FIGURE 8 Schematic diagram for calculating the angle between two skeletons.

3.4 | Human posture estimation and ergonomic risk assessment

Forward kinematics (Kucuk & Bingul, 2006) is adopted to recognize the posture of the target worker based on the detected keypoints. Specifically, as shown in Figure 7, starting at the root keypoint, forward kinematics computes the position of each body keypoint according to the position of its parent keypoint and relative rotation angle. Once the 3D posture is estimated, the vector of any skeleton can be obtained. Thus, the angle between any two skeletons can be calculated for the subsequent analysis. Figure 8a,b shows the vectors of the two skeletons; therefore, the angle between the two skeletons is computed as follows:

$$\theta = \arccos \left(\frac{a \cdot b}{|a| |b|} \right) \quad (4)$$



TABLE 1 Rapid Entire Body Assessment (REBA) levels (Hignett & McAtamney, 2000).

Action level	REBA score	Risk level	Action (including further assessment)
0	1	Negligible	None necessary
1	2–3	Low	May be necessary
2	4–7	Medium	Necessary
3	8–10	High	Necessary soon
4	11–15	Very high	Necessary now

Subsequently, based on the calculated angles, a widely used ergonomic risk assessment tool, REBA (Hignett & McAtamney, 2000), is adopted in this study. It has been proven that REBA can accurately and effectively assess ergonomic risk (Schwartz et al., 2019). Specifically, REBA first scores group A (i.e., trunk, neck, and legs) and group B (i.e., upper arms, lower arms, and wrist) according to the extension or flexion angles, and then integrates the scores of the two groups into an overall score. Finally, the REBA score can be obtained after adding the activity score (e.g., frequent movement). A higher REBA score represents a higher risk level. Table 1 lists specific REBA action levels (Hignett & McAtamney, 2000).

4 | EXPERIMENTS

4.1 | Datasets

4.1.1 | Open-source datasets

As mentioned in Section 3.1, two 2D human pose datasets (i.e., COCO and MPII) and two 3D human pose datasets (i.e., Human3.6 M and UP3D) are fused into a new dataset. COCO (Lin et al., 2014) is a large-scale object detection, segmentation, and captioning dataset that contains 250,000 people with keypoints. MPII (Andriluka et al., 2014) was collected from YouTube, including approximately 25,000 images that contain over 40,000 people with annotated keypoints. Note that an image may contain more than one person. Human3.6 M (Ionescu et al., 2013) is one of the most important public datasets for 3D body keypoint detection, which was captured from four calibrated cameras as a high-resolution 50-Hz video, containing 3.6 million 3D human poses and corresponding images of over 11 professional actors. UP3D (Lassner et al., 2017) was synthesized using the SMPLify method (Bogo et al., 2016) and contains high-quality 3D body model fitting for multiple human pose datasets. The attributes of the open-source datasets used in this study are concluded in Table 2.

TABLE 2 Overview of open-source datasets.

Dataset	Scene	# Keypoints	# Images (K)
COCO	Outdoor	17	150
MPII	Outdoor	16	25
Human3.6 M	Indoor	18	120
UP3D	Hybrid	24	7

4.1.2 | Newly established dataset

To further verify the accuracy and generalization of the proposed models, a validation dataset especially for workers in construction environments was established. The images were collected from the Internet and other datasets. Only 2D human poses of this dataset were annotated given the difficulty in annotating a 3D pose on a 2D image. This dataset contains 349 images and 616 people with annotated body keypoints. The dataset can be accessed at <https://drive.google.com/file/d/1jkQmhmJO9HbkcnpZlgIKTffX5NWCjIz/view?usp=sharing>.

4.2 | Experiment settings

To improve the generalization of the model, some data augmentation technologies were also used when training the model, including horizontal flip, rotation, occlusion, and color transformation. In particular, the occlusion augmentation stochastically occludes a small part of the images, which may make a few keypoints of the inside people invisible, forcing the HPE models to learn to estimate the positions of the occluded parts. This could enhance the HPE models' ability to capture 3D postures of the target people in 2D images with complicated scenes. The image size of the input is 256×192 , and the batch size is 128. The loss function used is Adam (Kingma & Ba, 2015), and the initial learning rate is taken as 0.001. The models were trained on a GPU Nvidia 3090 for 210 epochs, and the duration of the training process was approximately 5 days.

4.3 | Experiment results

4.3.1 | Training results

The training loss and testing accuracy are shown in Figure 9. The training datasets are listed in Table 2, and the testing dataset is the Human3.6 M validation set. Specifically, *FusionPose* and *FusionPose** exhibit good convergence. The training loss of *FusionPose** is lower than that of *FusionPose*. Correspondingly, the testing

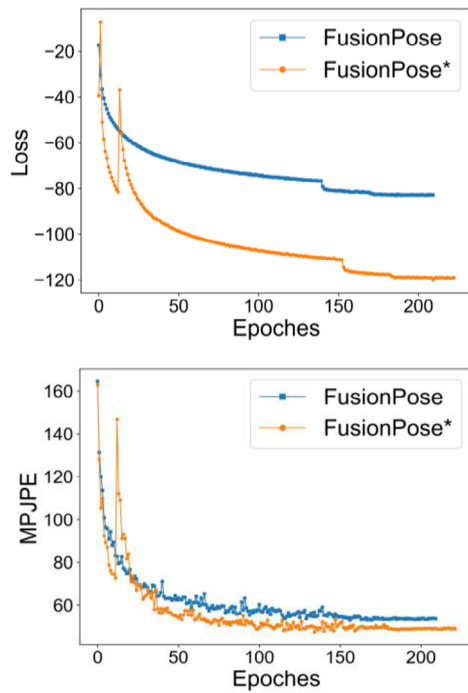


FIGURE 9 Training process.

accuracy of *FusionPose** is higher than that of *FusionPose*. Therefore, *FusionPose** can be adopted when a high assessment accuracy is required. However, it is noteworthy that *FusionPose** is slower than *FusionPose*. The computational cost of *FusionPose** is approximately 15 times that of *FusionPose*. Consequently, high-performance computational resources are required to ensure the real-time analysis of *FusionPose**, while the excellent efficiency of *FusionPose* makes it satisfy the requirement of real-time posture identification and risk assessment even with ordinary laptops. The specific recognition accuracy and computation speed of *FusionPose* and *FusionPose** will be compared in Sections 4.3.2, 4.3.3, and 5.

4.3.2 | Results on Human3.6 M

The two well-trained models were first tested on the Human3.6 M validation set. As introduced above, Human3.6 M is a 3D human pose dataset captured by four calibrated cameras and sensors in a laboratory. The Human3.6 M validation set contains approximately 8000 images. The proposed models were compared with other single-stage regression-based models that also use the annotated human box areas as the input. The corresponding results are listed in Table 3. Three key metrics, that is, the number of parameters, Giga floating-point operations (GFLOPs), and MPJPE, are reported. Specifically, the number of parameters is used to measure the size of

TABLE 3 Results on Human3.6 M validation set.

Methods	# Params (M)	GFLOPs	MPJPE
Sun et al. (2017)	–	–	59.1
Nibali et al. (2019)	57.9	29.3	49.5
Sun et al. (2018)	34.3	14.1	49.6
Moon et al. (2019)	34.3	14.1	53.3
Zhou et al. (2019)	49.6	41.7	47.7
J. Li et al. (2021)	23.8	5.4	48.6
<i>FusionPose</i>	3.0	0.34	53.2
<i>FusionPose*</i>	23.8	5.4	47.5

Abbreviations: GFLOPs, Giga floating-point operations; MPJPE, mean per joint position error.

models; GFLOPs are used to measure the computational cost of models; and MPJPE, the mean Euclidean distance between predicted joint positions and ground-truth joint positions, is used to measure the accuracy of the models.

As listed in Table 3, compared with Zhou et al. (2019), *FusionPose** realizes state-of-the-art performance with 0.2-mm improvement via much lower computation costs. Note that the model proposed by J. Li et al. (2021) features the same GFLOPs with *FusionPose**. Compared with J. Li et al. (2021), *FusionPose** achieves a 1.1-mm improvement in the MPJPE indicator. In contrast, *FusionPose* is the lightest model (with only 3.0 million parameters) among the listed studies and realizes comparable performance to *FusionPose** while reducing GFLOPs by 93.7%. Therefore, users can select the most suitable model from *FusionPose* and *FusionPose** according to the specific requirement. For example, *FusionPose** can be adopted if high accuracy is required, while *FusionPose* can be considered a good alternative when no high-performance device is available.

4.3.3 | Ablation study

Moreover, the ablation experiments were implemented based on the *FusionPose** model to demonstrate the effectiveness of the proposed co-learning method. As indicated by Table 4, four types of fused datasets are considered in the ablation study. Specifically, *FusionPose** trained on the fused dataset realizes the best performance via a 1.1-mm improvement over the model trained on the dataset that only contains Human3.6 M and MPII. This verifies the effectiveness of the proposed co-learning method. It is interesting to note that, compared with the dataset containing only Human3.6 M and MPII, the fused dataset consisting of Human3.6 M, MPII, and COCO does not improve the performance of the model on Human3.6 M validation set. This is because the scene of Human3.6 M



Learning only
from Human3.6M
and MPII

Co-learning from
the fused dataset

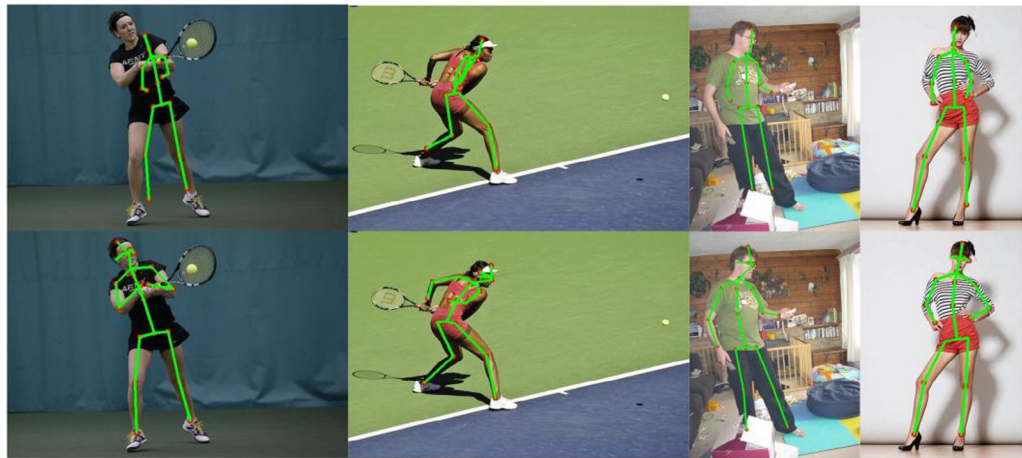


FIGURE 10 Qualitative comparison between the performances of *FusionPose** learning only from Human3.6 M and MPII and those co-learning from the fused dataset.

TABLE 4 Ablation study on Human3.6 M validation set.

Model	Datasets	MPJPE
<i>FusionPose</i> ^a	Human3.6 M, MPII	48.6
<i>FusionPose</i> ^a	Human3.6 M, MPII, COCO	50.2
<i>FusionPose</i> ^a	Human3.6 M (ds ^a), MPII, COCO	48.1
<i>FusionPose</i> ^a	Human3.6 M (ds ^a), MPII, COCO, UP3D (the fused dataset)	47.5

Abbreviation: MPJPE, mean per joint position error.

^aDown-sampling.

dataset is homogeneous, while that of the COCO dataset is diverse. Hence, the model can be overfitted when trained on these two datasets with similar sizes. When the downsampling is performed for the Human3.6 M dataset, the model trained on the fused dataset consisting of Human3.6 M, MPII, and COCO can focus on learning the 2D features from the diverse COCO and MPII datasets and 3D features from the simplified Human3.6 M dataset.

Furthermore, as shown in Figure 10, this study compared the performances of *FusionPose** learning only from Human3.6 M and MPII with those co-learning from the fused dataset. The former has a bad generalization and cannot accurately capture the postures of persons in images with complicated scenes since the model is trained on datasets with similar scenes. However, by co-learning from the fused dataset, *FusionPose** has learned human poses in various scenes and thus can accurately capture the persons' postures, even for those occluded parts, for example, the right arm in the second image and the right foot in the third image in Figure 10. These results are consistent with the stated theory in Section 3.1.

4.3.4 | Results on newly established dataset

In this section, the testing of the models using a new validation dataset annotated by ourselves is introduced. Owing to the difficulty in annotating 3D labels, only the 2D-pixel coordinates of the workers' body keypoints were annotated in the newly established dataset. Thus, the MPJPE result could not be calculated on the new dataset. In contrast, the *mAP* value that could be calculated by object keypoint similarity (OKS) is adopted to measure the accuracy of the models. OKS is calculated as follows:

$$OKS = \frac{\sum_i \exp(-d_i^2 / 2s^2 k_i^2) \delta(v_i > 0)}{\sum_i \delta(v_i > 0)} \quad (5)$$

where d_i represents the Euclidean distance between i th detected keypoint and ground truth, s the area of the human box, k_i the per-keypoint annotation variance, and v_i the visibility flag. The *mAP* value represents the mean of average precisions at different OKS thresholds (Lin et al., 2014). Generally, the *mAP* value greater than 70.0 represents the model performs well. *FusionPose* and *FusionPose** realize 76.0 and 82.8 of *mAP* on the new dataset, respectively, which demonstrates that the well-trained models are sufficiently robust to accurately capture workers' postures in construction environments.

The qualitative results of the newly established validation dataset are shown in Figure 11. The first two images are clear, in which the target workers' keypoints are not occluded. Thus, the HPE model can accurately capture the 3D postures of workers. Regarding the last two images, however, the workers inside are partially occluded, but the HPE model can still relatively accurately estimate the positions of the occluded parts. This is because a few images

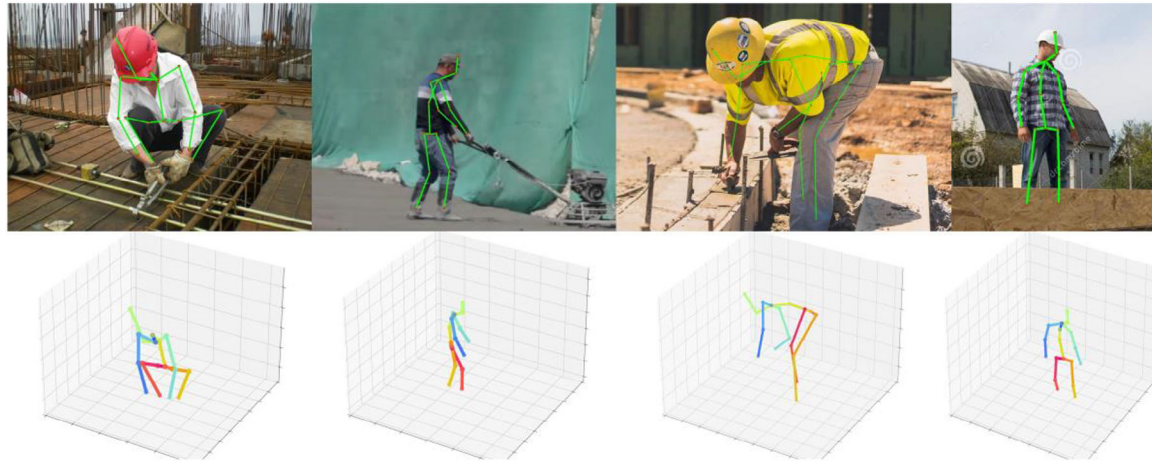


FIGURE 11 Qualitative results on the new dataset.

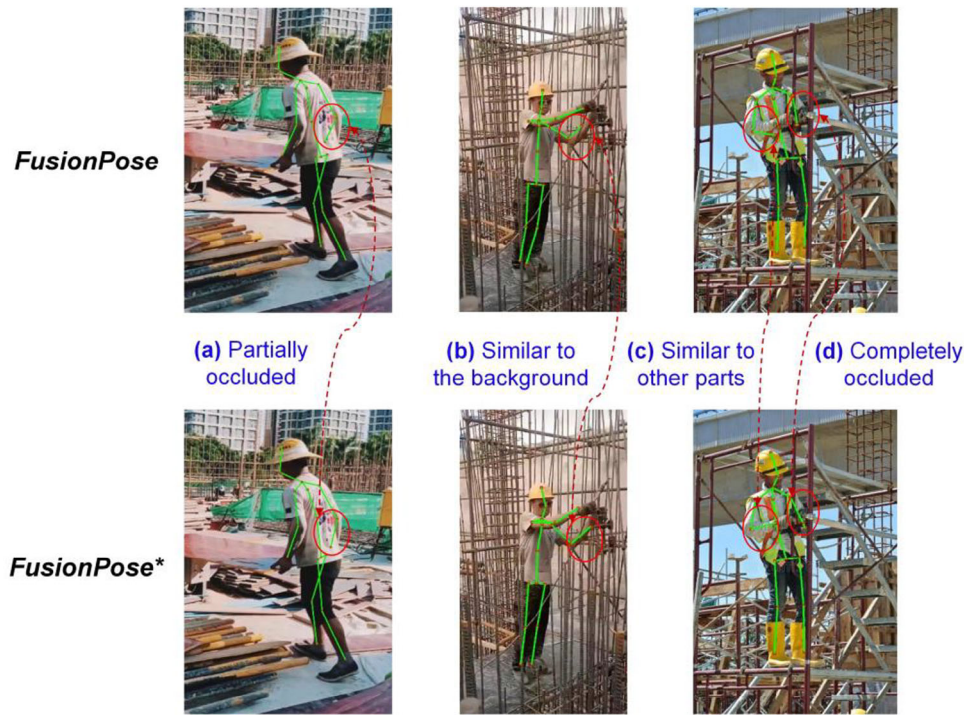


FIGURE 12 Qualitative comparison between *FusionPose* and *FusionPose** on the new dataset.

in the fused dataset contain partially occluded people, and the positions of the occluded parts are estimated by annotators when annotating the images. Thus, the HPE models can learn to do so. In addition, the dataset is augmented by occlusion, which furthermore enhances the HPE models' ability to predict the coordinates of occluded keypoints.

Furthermore, Figure 12 illustrates a qualitative comparison between *FusionPose* and *FusionPose** on the new dataset. It can be seen that both models can accurately capture workers' postures, whereas *FusionPose** performs better than *FusionPose* in detail. In particular, the right

forearm of the worker in Figure 12a is partially occluded by his body, and thus the position of the right wrist estimated by *FusionPose* is slightly different from the ground truth. However, *FusionPose** can still accurately estimate the position of the partially occluded part. In addition, since the color of the worker's skin is similar to the background in Figure 12b (especially the left forearm of the worker is quite similar to the rebars), the models may be confused. Similarly, the color of the worker's right forearm is similar to his body in Figure 12c, which makes it more difficult to accurately estimate the position of the



right wrist. In both circumstances, *FusionPose** performs better than *FusionPose*, that is, *FusionPose** can accurately capture the confusing parts, but *FusionPose* may get lost. However, if a part is completely occluded, for example, the left forearm of the worker in Figure 12d, both models can only roughly estimate the position of the part, as the annotators may not be sure where the specific position of the part is. Regarding the ambiguity of the keypoints' positions due to occlusion, the multi-view camera system may be an effective solution, which is not the scope of this study and will be discussed in Section 6. In conclusion, both models perform well when capturing clear body parts (e.g., no occlusion), while *FusionPose** performs better than *FusionPose* if a part is occluded by or similar to other parts.

5 | APPLICATION

The capability of the proposed models to accurately recognize the 3D postures of workers in complex construction environments was demonstrated in Section 4 based on various images. To further illustrate the feasibility of the proposed framework in the real-time ergonomic risk assessment, the well-trained HPE models were combined with the pose-tracking technology (see Section 3.3) and REBA tool (see Section 3.4) to capture the 3D postures of the workers in a video and identify the ergonomic risk in real time.

The video was photographed using a mobile phone during a real construction environment. The resolution is 1280×720 pixels. The proposed framework was first tested with an Nvidia RTX 3090 GPU. The average time required for *FusionPose*- and *FusionPose**-based methods to process a frame in the video is approximately 17 and 32 ms, respectively. This indicates that the proposed *FusionPose*- and *FusionPose**-based methods can realize real-time ergonomic risk assessment of 58- and 31-frame-per-second videos, where the frame rates are higher than ordinary cameras. In addition, the proposed method was also tested on an ordinary laptop equipped with 6G GPU resources. The average time required by *FusionPose*- and *FusionPose**-based methods to process a frame in the video is approximately 37 and 78 ms, respectively. Thus, even when no high-performance GPU resources are accessible, the *FusionPose*-based method can still address a 27-frame-per-second video, realizing the real-time ergonomic risk assessment for construction workers.

Examples of the risk assessment results for three typical construction operations are shown in Figure 13. First, the worker squats and deals with the reinforcement bars. The REBA score of the worker's posture is 9. Correspondingly,

the risk level is high, which implies that the worker cannot hold this posture for a long time. Then, the worker stands up and walks, making the risk level low. Finally, the worker is bent to check the reinforcement bars, and the REBA score of the posture is 4, representing a medium-level ergonomic risk.

The aforementioned test indicates that the *FusionPose**-based method can rationally assess the ergonomic risk to workers in real time provided that high-performance GPU resources are accessible. In contrast, when there is no high-performance GPU resource, the *FusionPose*-based method is a good alternative that realizes the tradeoff between accuracy and efficiency. Users can freely select the most suitable method from the two solutions according to the specific condition. All the test results of the video based on the *FusionPose*- and *FusionPose**-based methods can be found at <https://www.youtube.com/watch?v=gXTvXkOsZwo> and https://www.youtube.com/watch?v=x3DO_hJupo8, respectively.

6 | DISCUSSION

A posture-based ergonomic assessment is one of the most significant methods for measuring the physical exposure to WMSD risk of workers in construction environments. Nevertheless, to the best of the authors' knowledge, studies regarding assessing the ergonomic risks of workers in real time are rarely reported. In this study, a novel framework that can realize real-time assessment of workers' ergonomic risks based on co-learning, regression-based HPE model, pose-tracking technology, and REMA tool is proposed. Specifically, a co-learning method that can be used to train the 3D regression-based HPE model is proposed. The HPE models trained on the fused dataset can learn 2D information from diverse 2D human pose datasets as well as depth information from 3D human pose datasets, improving the accuracy level of recognizing 3D postures from 2D images. Subsequently, pose-tracking technology is adopted to detect the worker from video frames without using an object detection model for each frame, which drastically reduces the computational cost of the assessment process. Finally, HPE technology and an ergonomic risk assessment tool are adopted to quantify the ergonomic risk level of workers. The main advantages of the proposed assessment framework are as follows.

1. Powered by the proposed co-learning method, the framework features state-of-the-art accuracy on Human3.6 M validation set via significantly lower computational costs, compared with existing single-stage regression-based models. Moreover, the novel co-learning technology can improve the generalization



(a)



(b)



(c)

FIGURE 13 Real-time assessment results on videos photographed in a construction environment: (a) squatting and risk level is high, (b) walking and risk level is low, and (c) bending and risk level is medium.

of the model, enabling the model to accurately capture the 3D postures of the target worker from 2D images with complicated construction scenes.

- Based on the developed regression-based 3D HPE model and the adopted pose-tracking technology, the proposed method exhibits excellent efficiency with significantly fewer model parameters and lower computational costs, promoting its applicability in real construction projects. Even when no high-performance GPU resources are accessible, the proposed method can still realize the real-time ergonomic risk assessment for construction workers.
- The proposed framework provides two options, that is, *FusionPose*- and *FusionPose**-based methods, for users to choose according to their accessibility to high-performance computing resources. The former can be selected to realize high assessment accuracy when high-performance GPU resources are accessible, while the latter is a good alternative to realize the trade-off between accuracy and efficiency when there is no high-performance GPU resource.

Most construction sites have video monitoring systems, making the proposed method easy to apply. A potential application involves installing a small-size sensor in the worker's helmet that can receive ergonomic risk assessment results and alert the worker if he/she has held a high-risk posture for a long time. Another possible application is that the method can be used to count the ergonomic risk level of each worker daily to allocate and manage construction tasks reasonably. These applications are easy to realize at a low cost, but they are promising for drastically reducing workers' physical exposure to WMSD risks. Furthermore, it is noteworthy that the proposed method is not limited to construction workers. Nowadays, the physical exposure of employees engaged in jobs such as coding and clerical work to WMSD risks is becoming higher due to the social development pattern. The proposed method can also be applied to help these high-risk personnel reduce WMSD risks.

This study presents the following limitations, which can be addressed in the future.

- The method can only be utilized to assess the ergonomic risks of one worker captured by a single camera. In construction sites, there is usually more than one worker in the view of the camera.
- Another limitation is that the occlusion issue cannot be completely addressed by using the proposed method. A promising alternative for improving the recognition accuracy of occluded parts involves integrating vision information from multiple cameras so that all parts of each worker can be completely captured.

Nevertheless, the proposed framework can be applied to the assessment of workers' ergonomic risks with broad prospects. Multi-view camera systems will be considered to polish up the proposed method in the subsequent study. In addition, this study opens a few research perspectives for the future. First, this study can be further developed to assess the ergonomic risks of multi-workers from 2D images. In particular, an auxiliary model may be needed to estimate the absolute depth of workers (Shen et al., 2023), based on which the postures of all workers in 3D space and corresponding ergonomic risks can be obtained. Furthermore, the proposed framework can be adapted to other industries with high physical demand, for example, automobile manufacturing factories, reducing the possible ergonomic risks to workers. In addition, the proposed method can be integrated with other supervised machine learning/classification algorithms, for example, neural dynamic classification algorithm (Rafiei & Adeli, 2017), dynamic ensemble learning algorithm (Alam et al., 2020), finite element machine for fast learning (Pereira et al., 2020), and self-supervised learning (Rafiei et al., 2022), to realize a comprehensive assessment of the ergonomic risks to workers. For example, besides the awkward postures of workers, environmental factors such as noise and dust can also cause damage to workers, which can be recognized by an ensemble learning algorithm (Alam



et al., 2020) or classification algorithm (Rafiei & Adeli, 2017) when assessing the working risks to workers.

7 | CONCLUSION

It is crucial to monitor workers' postures and identify physical exposure to WMSD risks. In this study, a real-time assessment method for the ergonomic risks of workers in construction environments is proposed. The main conclusions are as follows.

1. The proposed co-learning method enables the HPE model to learn 2D and 3D human poses simultaneously from the fused dataset, ensuring that the HPE model can accurately recognize the 3D postures of workers directly from 2D images.
2. By co-learning from the fused dataset, the developed regression-based 3D HPE model, *FusionPose**, can realize state-of-the-art performance via 0.2-mm improvement on the Human3.6 M validation dataset with much lower computational costs, and the light-weight model, *FusionPose*, features comparable performance to *FusionPose** while reducing the GFLOPs by 93.7%.
3. With an Nvidia RTX 3090 GPU, the average time required for *FusionPose*- and *FusionPose**-based methods to process a frame in videos is approximately 17 and 32 ms, respectively, verifying that the proposed novel framework can realize a real-time assessment of workers' ergonomic risks in real construction environments.
4. The proposed framework provides two options for users to select according to their accessibility to high-performance computing resources. It is promising to apply the proposed method to drastically reduce the physical exposure risk to WMSDs for construction workers. Besides, it is possible for employees engaged in jobs such as coding and clerical work to benefit from the proposed technologies.

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REFERENCES

- Adeli, H., & Karim, A. (2001). *Construction scheduling, cost optimization and management*. CRC Press.
- Alam, K. M. R., Siddique, N., & Adeli, H. (2020). A dynamic ensemble learning algorithm for neural networks. *Neural Computing and Applications*, 32, 8675–8690.
- Andriluka, M., Pishchulin, L., Gehler, P., & Schiele, B. (2014). 2D human pose estimation: New benchmark and state of the art analysis. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, Columbus, OH (pp. 3686–3693).
- Bao, Y., Tang, Z., Li, H., & Zhang, Y. (2019). Computer vision and deep learning-based data anomaly detection method for structural health monitoring. *Structural Health Monitoring*, 18(2), 401–421.
- Bazarevsky, V., Grishchenko, I., Raveendran, K., Zhu, T., Zhang, F., & Grundmann, M. (2020). *BlazePose: On-device real-time body pose tracking*. arXiv preprint. arXiv:2006.10204.
- Bernardes, J. M., Gómez-Salgado, J., Ruiz-Frutos, C., & Dias, A. (2019). Self-reports of musculoskeletal symptoms as predictors of work-related accidents: A hospital-based case-control study. *Safety Science*, 115, 103–109.
- Bogo, F., Kanazawa, A., Lassner, C., Gehler, P., Romero, J., & Black, M. J. (2016). Keep it SMPL: Automatic estimation of 3D human pose and shape from a single image. *European Conference on Computer Vision*, Glasgow, UK (pp. 561–578).
- Boschman, J. S., van der Molen, H. F., Sluiter, J. K., & Frings-Dresen, M. H. (2012). Musculoskeletal disorders among construction workers: A one-year follow-up study. *BMC Musculoskeletal Disorders*, 13(1), 196.
- Center for Construction Research and Training (CPWR). (2018). *The construction chart book: The U.S. construction industry and its workers* (6th ed.). CPWR. https://www.cpwrt.com/sites/default/files/publications/The_6th_Edition_Construction_eChart_Book.pdf
- Chen, J., Qiu, J., & Ahn, C. (2017). Construction worker's awkward posture recognition through supervised motion tensor decomposition. *Automation in Construction*, 77, 67–81.
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2021). An image is worth 16 × 16 words: Transformers for image recognition at scale. *International Conference on Learning Representations*, Vienna, Austria (pp. 1–21).
- Duan, R., Deng, H., Tian, M., Deng, Y., & Lin, J. (2022). SODA: A large-scale open site object detection dataset for deep learning in construction. *Automation in Construction*, 142, 104499.
- Girshick, R. (2015). Fast R-CNN. *Proceedings of the IEEE International Conference on Computer Vision*, Santiago, Chile (pp. 1440–1448).
- Golparvar-Fard, M., Heydarian, A., & Niebles, J. C. (2013). Vision-based action recognition of earthmoving equipment using spatio-temporal features and support vector machine classifiers. *Advanced Engineering Informatics*, 27(4), 652–663.
- Guo, H., Lin, J. R., & Yu, Y. (2023). Intelligent and computer technologies' application in construction. *Buildings*, 13(3), 641.
- Harichandran, A., Raphael, B., & Mukherjee, A. (2023). Equipment activity recognition and early fault detection in automated construction through a hybrid machine learning framework. *Computer-Aided Civil and Infrastructure Engineering*, 38(2), 253–268.
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, Las Vegas, NV (pp. 770–778).
- Hignett, S., & McAtamney, L. (2000). Rapid entire body assessment (REBA). *Applied Ergonomics*, 31(2), 201–205.
- Howard, A. G., Zhu, M., Chen, B., Kalenichenko, D., Wang, W., Weyand, T., Andreetto, M., & Adam, H. (2017). *MobileNets:*



- Efficient convolutional neural networks for mobile vision applications.* arXiv preprint. arXiv:1704.04861.
- Ionescu, C., Papava, D., Olaru, V., & Sminchisescu, C. (2013). Human3.6M: Large scale datasets and predictive methods for 3D human sensing in natural environments. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 36(7), 1325–1339.
- Janowitz, I. L., Gillen, M., Ryan, G., Rempel, D., Trupin, L., Swig, L., Mullen, K., Rugulies, R., & Blanc, P. D. (2006). Measuring the physical demands of work in hospital settings: Design and implementation of an ergonomics assessment. *Applied Ergonomics*, 37(5), 641–658.
- Ji, X., Miao, Z., & Kromanis, R. (2020). Vision-based measurements of deformations and cracks for RC structure tests. *Engineering Structures*, 212, 110508.
- Johansen, K. W., Schultz, C., & Teizer, J. (2022). Hazard ontology and 4D benchmark model for facilitation of automated construction safety requirement analysis. *Computer-Aided Civil and Infrastructure Engineering*, 38, 2128–2144.
- Jung, S., Jeoung, J., Lee, D. E., Jang, H., & Hong, T. (2023). Visual-auditory learning network for construction equipment action detection. *Computer-Aided Civil and Infrastructure Engineering*, 38, 1916–1934.
- Karim, A., & Adeli, H. (1999a). OO information model for construction project management. *Journal of Construction Engineering and Management*, 125(5), 361–367.
- Karim, A., & Adeli, H. (1999b). CONSCOM: An OO construction scheduling and change management system. *Journal of Construction Engineering and Management*, 125(5), 368–376.
- Keyserling, W. M., Brouwer, M., & Silverstein, B. A. (1993). The effectiveness of a joint labor-management program in controlling awkward postures of the trunk, neck, and shoulders: Results of a field study. *International Journal of Industrial Ergonomics*, 11(1), 51–65.
- Kingma, D. P., & Ba, J. L. (2015). Adam: A method for stochastic gradient descent. *International Conference on Learning Representations*, San Diego, USA (pp. 1–15).
- Kucuk, S., & Bingul, Z. (2006). *Robot kinematics: Forward and inverse kinematics*. INTECH Open Access Publisher.
- Lassner, C., Romero, J., Kiefel, M., Bogo, F., Black, M. J., & Gehler, P. V. (2017). Unite the people: Closing the loop between 3D and 2D human representations. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, Honolulu, HI (pp. 6050–6059).
- Li, C., & Lee, G. H. (2019). Generating multiple hypotheses for 3d human pose estimation with mixture density network. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Long Beach, CA (pp. 9887–9895).
- Li, J., Bian, S., Zeng, A., Wang, C., Pang, B., Liu, W., & Lu, C. (2021). Human pose regression with residual log-likelihood estimation. *Proceedings of the IEEE/CVF International Conference on Computer Vision*, Montreal, BC, Canada (pp. 11025–11034).
- Lin, T. Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., & Zitnick, C. L. (2014). Microsoft COCO: Common objects in context. In D. Fleet, T. Pajdla, B. Schiele, & T. Tuytelaars (Eds.), *European conference on computer vision. ECCV 2014. Lecture notes in computer science*, (Vol. 8693, pp. 740–755). Springer.
- Luo, X., Li, H., Yang, X., Yu, Y., & Cao, D. (2019). Capturing and understanding workers' activities in far-field surveillance videos with deep action recognition and Bayesian nonparametric learning. *Computer-Aided Civil and Infrastructure Engineering*, 34(4), 333–351.
- Li, Y., Yang, S., Liu, P., Zhang, S., Wang, Y., Wang, Z., Yang, W., & Xia, S. T. (2022). Simcc: A simple coordinate classification perspective for human pose estimation. In *European Conference on Computer Vision*, Cham: Springer Nature Switzerland (pp. 89–106).
- Luo, X., Li, H., Yu, Y., Zhou, C., & Cao, D. (2020). Combining deep features and activity context to improve recognition of activities of workers in groups. *Computer-Aided Civil and Infrastructure Engineering*, 35(9), 965–978.
- Mao, W., Ge, Y., Shen, C., Tian, Z., Wang, X., Wang, Z., & den Hengel, A. V. (2022). Poseur: Direct human pose regression with transformers. In S. Avidan, G. Brostow, M. Cissé, G. M. Farinella, & T. Hassner (Eds.), *European conference on computer vision. Lecture notes in computer science* (Vol. 13666, pp. 72–88). Springer.
- Martínez-Rojas, M., Marín, N., & Vila, M. A. (2015). An approach for the automatic classification of work descriptions in construction projects. *Computer-Aided Civil and Infrastructure Engineering*, 30(12), 919–934.
- Moon, G., Chang, J. Y., & Lee, K. M. (2019). Camera distance-aware top-down approach for 3D multi-person pose estimation from a single RGB image. *Proceedings of the IEEE/CVF International Conference on Computer Vision*, Seoul, Korea (pp. 10133–10142).
- Newell, A., Yang, K., & Deng, J. (2016). Stacked hourglass networks for human pose estimation. In B. Leibe, J. Matas, N. Sebe, & M. Welling (Eds.), *European conference on computer vision. Lecture notes in computer science* (Vol. 9912, pp. 483–499). Springer.
- Nibali, A., He, Z., Morgan, S., & Prendergast, L. (2019). 3D human pose estimation with 2D marginal heatmaps. *2019 IEEE Winter Conference on Applications of Computer Vision (WACV)*, Waikoloa Village, HI (pp. 1477–1485).
- Occupational Safety and Health Branch (OSH). (2017). *Occupational safety and health statistics 2017*. Labour Department, The Government of the Hong Kong Special Administrative Region. https://www.labour.gov.hk/eng/osh/pdf/OSH_Statistics_2017_EN.pdf
- Pereira, D. R., Piteri, M. A., Souza, A. N., Papa, J. P., & Adeli, H. (2020). FEMa: A finite element machine for fast learning. *Neural Computing and Applications*, 32, 6393–6404.
- Rafiei, M. H., & Adeli, H. (2017). A new neural dynamic classification algorithm. *IEEE Transactions on Neural Networks and Learning Systems*, 28(12), 3074–3083.
- Rafiei, M. H., Gauthier, L. V., Adeli, H., & Takabi, D. (2022). Self-supervised learning for electroencephalography. *IEEE Transactions on Neural Networks and Learning Systems*. Advance online publication. [10.1109/TNNLS.2022.3190448](https://doi.org/10.1109/TNNLS.2022.3190448)
- Ray, S. J., & Teizer, J. (2012). Real-time construction worker posture analysis for ergonomics training. *Advanced Engineering Informatics*, 26(2), 439–455.
- Ren, B., Luo, X., Li, H., Chen, J., & Wang, Y. (2022). Gait trajectory-based interactive controller for lower limb exoskeletons for construction workers. *Computer-Aided Civil and Infrastructure Engineering*, 37(5), 558–572.
- Ren, S., He, K., Girshick, R., & Sun, J. (2015). Faster R-CNN: Towards real-time object detection with region proposal networks. *Advances in Neural Information Processing Systems*, 28, Montreal, Quebec, Canada.
- Roberts, D., Torres Calderon, W., Tang, S., & Golparvar-Fard, M. (2020). Vision-based construction worker activity analysis



- informed by body posture. *Journal of Computing in Civil Engineering*, 34(4), 04020017.
- Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L. C. (2018). MobileNetV2: Inverted residuals and linear bottlenecks. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, Salt Lake City, UT (pp. 4510–4520).
- Seo, J., & Lee, S. (2021). Automated postural ergonomic risk assessment using vision-based posture classification. *Automation in Construction*, 128, 103725.
- Schwartz, A. H., Albin, T. J., & Gerberich, S. G. (2019). Intra-rater and inter-rater reliability of the rapid entire body assessment (REBA) tool. *International Journal of Industrial Ergonomics*, 71, 111–116.
- Shen, J., Yan, W., Qin, S., & Zheng, X. (2023). A self-supervised monocular depth estimation model with scale recovery and transfer learning for construction scene analysis. *Computer-Aided Civil and Infrastructure Engineering*, 38, 1142–1161.
- Sun, X., Shang, J., Liang, S., & Wei, Y. (2017). Compositional human pose regression. *Proceedings of the IEEE International Conference on Computer Vision*, Venice, Italy (pp. 2602–2611).
- Sun, X., Xiao, B., Wei, F., Liang, S., & Wei, Y. (2018). Integral human pose regression. *Proceedings of the European Conference on Computer Vision (ECCV)*, Munich, Germany (pp. 529–545).
- Tang, S., & Golparvar-Fard, M. (2021). Machine learning-based risk analysis for construction worker safety from ubiquitous site photos and videos. *Journal of Computing in Civil Engineering*, 35(6), 04021020.
- Wang, D., Dai, F., & Ning, X. (2015). Risk assessment of work-related musculoskeletal disorders in construction: State-of-the-art review. *Journal of Construction Engineering and Management*, 141(6), 04015008.
- Wang, J., Chen, D., Zhu, M., & Sun, Y. (2021). Risk assessment for musculoskeletal disorders based on the characteristics of work posture. *Automation in Construction*, 131, 103921.
- Wang, J., Tan, S., Zhen, X., Xu, S., Zheng, F., He, Z., & Shao, L. (2021). Deep 3D human pose estimation: A review. *Computer Vision and Image Understanding*, 210, 103225.
- Wu, C., Li, X., Jiang, R., Guo, Y., Wang, J., & Yang, Z. (2023). Graph-based deep learning model for knowledge base completion in constraint management of construction projects. *Computer-Aided Civil and Infrastructure Engineering*, 38(6), 702–719.
- Xiao, B., Wu, H., & Wei, Y. (2018). Simple baselines for human pose estimation and tracking. *Proceedings of the European Conference on Computer Vision (ECCV)*, Munich, Germany (pp. 466–481).
- Yan, X., Li, H., Li, A. R., & Zhang, H. (2017). Wearable IMU-based real-time motion warning system for construction workers' musculoskeletal disorders prevention. *Automation in Construction*, 74, 2–11.
- Yan, X., Zhang, H., & Zhang, W. (2023). Intelligent monitoring and evaluation for the prefabricated construction schedule. *Computer-Aided Civil and Infrastructure Engineering*, 38(3), 391–407.
- Yoong, N. K. M., Perring, J., & Mobbs, R. J. (2019). Commercial postural devices: A review. *Sensors*, 19(23), 5128.
- Yan, X., Li, H., Wang, C., Seo, J., Zhang, H., & Wang, H. (2017). Development of ergonomic posture recognition technique based on 2D ordinary camera for construction hazard prevention through view-invariant features in 2D skeleton motion. *Advanced Engineering Informatics*, 34, 152–163.
- Yu, Y., Li, H., Yang, X., Kong, L., Luo, X., & Wong, A. Y. (2019). An automatic and non-invasive physical fatigue assessment method for construction workers. *Automation in Construction*, 103, 1–12.
- Yu, Y., Umer, W., Yang, X., & Antwi-Afari, M. F. (2021). Posture-related data collection methods for construction workers: A review. *Automation in Construction*, 124, 103538.
- Zhang, H., Yan, X., & Li, H. (2018). Ergonomic posture recognition using 3D view-invariant features from single ordinary camera. *Automation in Construction*, 94, 1–10.
- Zhao, J., & Obonyo, E. (2020). Convolutional long short-term memory model for recognizing construction workers' postures from wearable inertial measurement units. *Advanced Engineering Informatics*, 46, 101177.
- Zhao, J., & Obonyo, E. (2021). Applying incremental Deep neural networks-based posture recognition model for ergonomics risk assessment in construction. *Advanced Engineering Informatics*, 50, 101374.
- Zhou, K., Han, X., Jiang, N., Jia, K., & Lu, J. (2019). HEMlets pose: Learning part-centric heatmap triplets for accurate 3D human pose estimation. *Proceedings of the IEEE/CVF International Conference on Computer Vision*, Seoul, Korea (pp. 2344–2353).

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